

Scientific Vita

(Wissenschaftlicher Lebenslauf)

Christoph Gerhardt

born July 17, 1990 in Ilmenau, Germany

Education

- Oct 2019 – May 2026 (expected) Dr.-Ing. (PhD) in Computer Science**
Technische Universität Ilmenau, Faculty of Computer Science and Automation
Dissertation: "Comprehensive Outdoor Scene Understanding Through Multi-Scale Neural Networks: From 360° Object Detection to Panoramic Segmentation and Weather-Robust Augmentation"
Supervisor: Prof. Dr. rer. nat. Wolfgang Broll
Research Group: Virtual Worlds and Digital Games (VWDG)
- Oct 2014 – Sep 2017 M.Sc. Computer Engineering (Ingenieurinformatik)**
Technische Universität Ilmenau — Final Grade: 1.5 (very good)
Specialisation: Multimedia Information and Communication Systems.
Thesis: "Selektive Verschlüsselung von Gesichtern in Videodaten unter Verwendung des H.264-Standards"
- Oct 2010 – Sep 2014 B.Sc. Computer Engineering (Ingenieurinformatik)**
Technische Universität Ilmenau — Final Grade: 1.8 (good)
Thesis: "Entwicklung eines Verfahrens zur Verbesserung von Mikrofonklassifizierung basierend auf der Analyse von Umwelteinflüssen"
- Jun 2009 Abitur**
Gymnasium "Am Lindenberg", Ilmenau — Grade: 1.5

Professional Experience

- Feb 2019 – present Research Associate (Wissenschaftlicher Mitarbeiter)**
Technische Universität Ilmenau, Fachgebiet Virtuelle Welten und Digitale Spiele
Doctoral research on outdoor scene understanding with neural networks; development of datasets, architectures, and benchmarks; teaching; student supervision

- Feb 2017 – Jan 2019 Software Developer**
 Fraunhofer Institute for Digital Media Technology (IDMT), Ilmenau
Group: Virtual Acoustics — C++ and Python development for spatial audio rendering
- Jul 2016 – Jan 2017 Student Research Assistant (Hilfswiss. Mitarbeiter)**
 Fraunhofer IDMT, Ilmenau
Group: Media Distribution and Security — Face detection and selective encryption in H.264 video

Research Activities

Dissertation Research (2019–2026)

My doctoral research addresses the challenge of **comprehensive outdoor scene understanding** through a progression of increasingly complex perception tasks:

1. **360° Traffic Sign Detection** — Adapted object detection networks for equirectangular panoramic images; created a dataset of 2 500 images with 81 traffic sign classes; deployed in NOVASIB GmbH's TT-SIB INFOSYS® road inventory system (IEEE ITSC 2020).
2. **Universal Outdoor Scene Segmentation (OUTSIDE)** — Designed a multi-scale semantic segmentation architecture for unconstrained outdoor scenes; curated the OUTSIDE15k dataset (15 000 images, 24 classes) (IEEE MMSP 2021).
3. **Sky and Cloud Segmentation (SkyCloud)** — Developed SkyCloudNet for fine-grained sky and cloud segmentation from perspective images; released dataset and pre-trained models (IEEE ICIVC 2023).
4. **Panoramic Sky Segmentation (SkyCloud360)** — Extended sky/cloud segmentation to equirectangular 360° imagery; created a 600-image panoramic dataset with addressing of projection distortions (IEEE ICIVC 2025).
5. **Weather-Robust Augmentation (AWARE)** — Developed a three-pipeline benchmark framework (SWIFT, PRISM, PROVE) evaluating 25 augmentation strategies for weather robustness; generated the AWACS dataset (71 400 images) (IEEE Access, under review 2025).

Collaborative Research

In addition to the core dissertation work, I contributed to collaborative research within the VWDG group on:

- **Avatar and Agent Realism** — Behavioural realism of animated avatars in AR and VR environments, including single-point IK animation and full-body motion capture comparisons (IEEE VR 2024, Int. Conf. Cyberworlds 2024, The Visual Computer 2025).
- **AR Communication Systems** — User studies with older adults using wearable AR communication systems (ACM IMX 2024, ITAP/HCI 2025).
- **Gaze and Audiovisual Coherence** — Analysis of gaze behaviour and effects of visual realism and audio presentation in AR storytelling (ACM ETRA 2024, IEEE VR 2024).
- **Systematic Reviews** — Co-authored a systematic review on avatar and agent visualisation in HMD-based AR & VR (IEEE TVCG 2023).
- **Telepresence Quality** — Impact of partner representation (robot, avatar, human) on communication experience (ACM CSCW 2025).

Open-Source Contributions & Datasets

- **AWARE Framework:** SWIFT + PRISM + PROVE pipelines (<https://carhartt21.github.io/AWARE>)
- **AWACS Dataset:** 71 400 weather-augmented images
- **SkyCloud360:** 600 equirectangular images (<https://github.com/carhartt21/SkyCloud360>)
- **SkyCloudNet:** Sky/cloud segmentation models (<https://github.com/carhartt21/SkyCloudNet>)
- **OUTSIDE15k:** 15 000 images, 24 classes (<https://github.com/carhartt21/outsideNet>)
- **360° Detection:** 2 500 panoramic images, 81 sign classes (<https://github.com/carhartt21/360-Object-Detection>)

Conference Presentations

1. **Oral Presentation:** “SkyCloud360: Sky and Cloud Segmentation in Equirectangular Images,” 10th Int. Conf. Image, Vision and Computing (ICIVC), Chengdu, China, June 2025.
2. **Oral Presentation:** “SkyCloud: Neural Network-Based Sky and Cloud Segmentation from Natural Images,” 8th Int. Conf. Image, Vision and Computing (ICIVC), Dalian, China, August 2023.
3. **Oral Presentation:** “OUTSIDE: Multi-Scale Semantic Segmentation of Universal Outdoor Scenes,” IEEE 23rd Int. Workshop Multimedia Signal Processing (MMSP), Tampere, Finland, October 2021.
4. **Oral Presentation:** “Neural Network-Based Traffic Sign Recognition in 360° Images for Semi-Automatic Road Maintenance Inventory,” IEEE 23rd Int. Conf. Intelligent

Transportation Systems (ITSC), virtual, September 2020.

Teaching Activities

Courses (Teaching Assistant)

WS 2019–2025	Virtual and Augmented Reality Technische Universität Ilmenau Practical sessions on VR/AR fundamentals, 3D rendering, tracking systems, and immersive interaction design. Student project mentoring and grading.
SS 2020–2025	Virtuelle Welten Technische Universität Ilmenau Practical sessions on virtual worlds, real-time rendering, game engine development, and 3D modelling. Semester project supervision.
Ongoing	Computer Vision & Deep Learning Seminars Technische Universität Ilmenau Seminar supervision on semantic segmentation, object detection, weather-robust perception, and neural network architectures.

Thesis Supervision

Co-supervision of 4+ Bachelor's and Master's theses (2020–present) in the areas of:

- Semantic segmentation for outdoor environments
- Weather-robust perception and data augmentation
- Panoramic and 360° image analysis
- Avatar rendering and VR/AR applications

Funding

2019 – 2024	Research funded by the Free State of Thuringia (Grant No. 2017 FE 9147), co-financed by the European Union under the European Regional Development Fund (ERDF). Project focus: intelligent road infrastructure analysis using neural networks and 360° imagery.
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Industry Collaboration

2019 – 2022	Collaboration with NOVASIB GmbH on the integration of neural-network-based traffic sign detection into the TT-SIB INFOSYS® road maintenance inventory management system, enabling semi-automatic road infrastructure assessment from 360° panoramic imagery.
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Technical Skills

Programming	Python, C/C++/C#, Java, MATLAB/Octave, SQL, Bash, L ^A T _E X
ML & DL	PyTorch, TensorFlow/Keras, scikit-learn, OpenCV, ONNX, Albumentations, timm
Infrastructure	Git, Docker, Linux, Slurm HPC, Weights & Biases, Jupyter, CI/CD
Domains	Semantic Segmentation, Object Detection, Data Augmentation, 360° Vision, VR/AR

Languages

German	Native speaker
English	Fluent (C1+) — academic writing, conference presentations
French	Basic knowledge (A2)
Serbian	Basic knowledge (A2)

Certifications

2023	C-Lizenz (DFB Football Coaching License)
2023	EU Remote Pilot Certificate (A1/A3) — Drone operation, EU Reg. 2019/947

Academic Service

Reviewing	Ad-hoc reviewer for IEEE and ACM conferences
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Ilmenau, March 16, 2026